

# Data Dictionary — EuroCrop Agricultural Logistics Dataset

The dataset contains **53,305 records and 29 columns**. It includes information about crops, vehicles, transportation, storage, environmental conditions, costs, IoT sensor readings, inventory, spoilage, and logistics efficiency.

| Column Name           | Data Type | Description                                                      |
|-----------------------|-----------|------------------------------------------------------------------|
| Unnamed: 0            | Object    | Timestamp or record identifier associated with each observation. |
| Vehicle_Type          | Object    | Type/category of vehicle used for agricultural transportation.   |
| Crop_Type             | Object    | Type of agricultural crop being transported or handled.          |
| Harvest_Date          | Object    | Date on which the crop was harvested.                            |
| Crop_Yield            | Float     | Quantity or yield of crop produced/handled.                      |
| Storage_Temperature   | Float     | Temperature maintained in the crop storage facility.             |
| Storage_Humidity      | Float     | Humidity level inside the storage facility.                      |
| Fuel_Consumption      | Float     | Amount of fuel consumed during transportation or operations.     |
| Route_Distance        | Float     | Distance covered by the vehicle during transportation.           |
| Delivery_Time         | Float     | Time taken to complete the agricultural delivery.                |
| Traffic_Level         | Float     | Traffic-related factor affecting transportation operations.      |
| Temperature           | Float     | Environmental temperature during the logistics operation.        |
| Humidity              | Float     | Environmental humidity level during the operation.               |
| Vehicle_Load_Capacity | Float     | Maximum load capacity of the transportation vehicle.             |

| Column Name                    | Data Type | Description                                                                                     |
|--------------------------------|-----------|-------------------------------------------------------------------------------------------------|
| Vibration_Level                | Float     | Vibration level experienced during transportation.                                              |
| Queue_Time                     | Float     | Time spent waiting or queuing at logistics facilities/stations.                                 |
| Weather_Impact                 | Float     | Numerical measure representing the effect of weather conditions on logistics.                   |
| Station_Capacity               | Float     | Capacity of the logistics/transportation station.                                               |
| Operational_Cost               | Float     | Total operational cost associated with the logistics activity.                                  |
| Energy_Consumption             | Float     | Energy consumed during logistics operations.                                                    |
| IoT_Sensor_Reading_Temperature | Float     | Temperature reading collected through IoT sensors.                                              |
| IoT_Sensor_Reading_Humidity    | Float     | Humidity reading collected through IoT sensors.                                                 |
| IoT_Sensor_Reading_Light       | Float     | Light-level reading collected through IoT sensors.                                              |
| Warehouse_Storage_Time         | Float     | Amount of time crops remain stored in the warehouse.                                            |
| Inventory_Levels               | Float     | Quantity of agricultural inventory available at the relevant location.                          |
| Fuel_Costs                     | Float     | Cost incurred for fuel used in logistics operations.                                            |
| Spoilage_Risk                  | Float     | Measure indicating the potential risk of crop spoilage.                                         |
| Efficiency_Ratio               | Float     | Measure representing the efficiency of the logistics operation.                                 |
| Quality_Maintenance_Ratio      | Float     | Measure indicating how effectively crop quality is maintained throughout the logistics process. |

## **Data Dictionary Categories**

You can also group the columns like this in your documentation:

### **1. Crop Information**

- Crop\_Type
- Crop\_Yield
- Harvest\_Date

### **2. Vehicle & Transportation**

- Vehicle\_Type
- Vehicle\_Load\_Capacity
- Fuel\_Consumption
- Route\_Distance
- Delivery\_Time
- Traffic\_Level
- Vibration\_Level
- Queue\_Time

### **3. Storage & Inventory**

- Storage\_Temperature
- Storage\_Humidity
- Warehouse\_Storage\_Time
- Inventory\_Levels
- Station\_Capacity

### **4. Environmental Conditions**

- Temperature
- Humidity
- Weather\_Impact

### **5. IoT Sensor Data**

- IoT\_Sensor\_Reading\_Temperature
- IoT\_Sensor\_Reading\_Humidity
- IoT\_Sensor\_Reading\_Light

## **6. Cost & Energy**

- Operational\_Cost
- Energy\_Consumption
- Fuel\_Costs

## **7. Quality & Performance**

- Spoilage\_Risk
- Efficiency\_Ratio
- Quality\_Maintenance\_Ratio